

Statistical Studies of Pulsar Transverse Velocities and the Nulling Phenomena

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1 Pulsars : An Introduction

”Pulsars exhibit some of the most extreme conditions found anywhere in the Universe. They offer the opportunity to study physics in regimes unattainable in any terrestrial laboratory, and provide a powerful probe for exploring the interstellar medium.”

-Andrew G. Lyne and Francis Graham-Smith, Pulsar Astronomy

1.1 Formation of neutron stars

When a star runs out of its nuclear fuel, it “dies”. A dead star cannot support itself against gravitational collapse by generating thermal pressure. This end state of a star leads to the formation of compact objects. Depending on the mass of the star (M), it may end up as a white dwarf ($M \leq 1.4M_{\odot}$)¹, a neutron star or a black hole. Neutron stars and black holes originate from more massive stars. However, the dividing line between those stars that form neutron stars and those that form black holes is very uncertain because the final stages of evolution of massive stars are poorly understood[1]. Typical neutron stars have masses in the range $1.4\text{--}3M_{\odot}$. As the progenitor star collapses, the Fermi energy of the electrons increases to the point where it is energetically favorable for them to combine with protons to produce neutrons (via inverse beta decay), thus, resulting in an extremely compact star composed of nuclear matter. If the remnant star has a mass $M > 3M_{\odot}$ it continues collapsing to form a black hole.

1.2 Neutron star properties

Once formed, neutron stars no longer actively generate heat, and cool over time. They may, however, evolve through collision or accretion. They are supported against further collapse by neutron degeneracy pressure.

Neutron stars measure about 20km in diameter and are some of the most dense objects in the universe. The average density of a neutron star is $\sim 10^{14}\text{--}10^{15}\text{g/cc}$, close to or larger than the nuclear density. They exhibit extreme surface magnetic fields ($B \sim 10^8\text{--}10^{15}\text{G}$) and rapid rotation about their axes with spin periods ranging from $10^{-3}\text{--}10^2\text{s}$.

1.3 Identifying pulsars with neutron stars

There are objects which are known to emit pulses of radiation predominantly in the radio (and also at other) wavelengths. These objects are known as pulsars because of the periodic emissions. A pulsar can effectively be approximated as a magnetic dipole, with negligible contribution from the other magnetic moments. The magnetic axis is not typically aligned with the rotational axis. The pulses can be attributed to the passage of the radiation beam from the magnetic poles of the pulsar across the line of sight of the observer, like a lighthouse. Pulse periods of radio pulsars can be measured to a very high accuracy of about 1 part in 10^{16} . Although radio pulsars rotate at very stable speeds, they are observed to slow down at a very nominal rate over time as they emit radiation and lose their energy.

Only a few months before radio pulsars were discovered by Hewish et. al. (1968) [2], Franco Pacini had already published a paper (Pacini 1967 [3]) where he showed that a rapidly rotating neutron star, with a strong dipolar field, would act as a very energetic electric generator which could provide a source of energy for radiation from a surrounding nebula (such as the Crab nebula).

¹ $M_{\odot}$ = Solar mass = $1.989 \times 10^{33}\text{g}$

Subsequent observational facts ruled out all other possibilities of pulsars being something apart from rotating neutron stars.

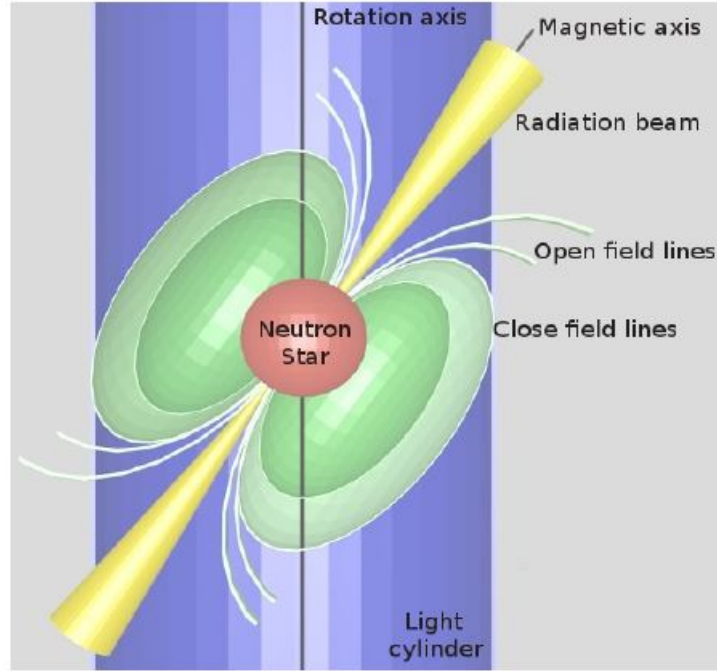


Figure 1: The schematic diagram of a radio pulsar to illustrate various regions around the neutron star (not to scale). The field lines enclosing inside the light cylinder are classified the close field line region while the field lines which cross the light cylinder belong to open field line region[4].

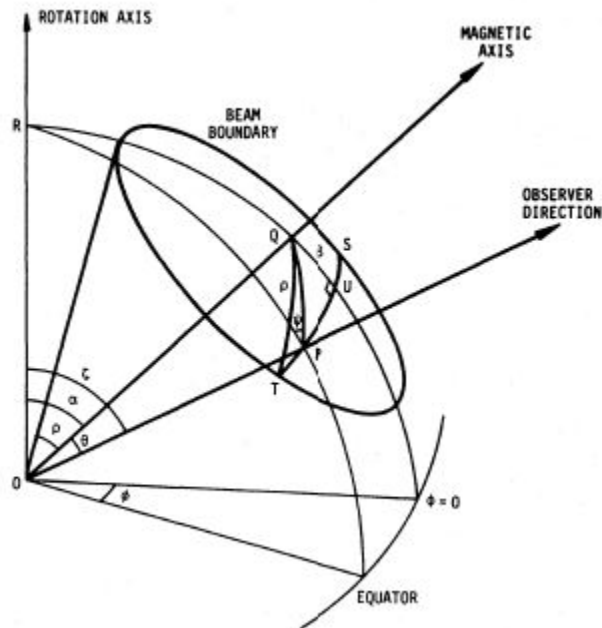


Figure 2: Diagram of pulsar geometry showing the angle α between rotation and magnetic axis along with other standard angle labels.[5]

2 Nulling

2.1 Radio emission

Extensive research have been focused on developing the emission mechanism model which can give rise to observed phenomena in radio pulsars. The main interest in pulsar phenomenology is the mechanism to generate the radio waves, with the brightness temperature of around 10^{25} – 10^{31} K. Such mechanism has to be coherent, as incoherent phenomena cannot account for such bright radiation. Among the various theories, the *two stream instability* to cause bunching is the most favored one. Positrons which escape the gap on top of the polar cap, acquire relativistic velocities with Lorentz factor, γ , reaching up to 10^6 , which are also known as the *primary particles*. These high energy primary particles do not get further accelerated beyond the gap as they enter the region of the magnetosphere where $E \cdot B = 0$. High energy photons are emitted when they travel in the curved magnetic field lines, through curvature radiation. A few of these photons again interact with the magnetic field to produce further pairs of $e^- - e^+$, which are known as the *secondary particles*. However, these secondary particles have much lower energies (with γ of around 800). In the absence of any accelerating electric field, both secondary particles pair travel in the outward direction, along the curved field line to conserve momentum. These particles are the primary source in three different variants of two-stream instability theories which can give rise to the observed radio emission, viz. (a) interaction between primary and secondary particles [6], (b) electrostatic interaction between electron and positron of the secondary particles [7], (c) interaction between clouds of secondary plasma [8][9][10].

2.2 Pulse nulling

A set of radio pulsars exhibit abrupt cessation of pulsed emission for an interval of time, after which they resume normal pulsed emission. This phenomenon is termed as ‘nulling’. Essentially a null means that the pulse energy decreases by a large factor over one or two pulse periods and then increases just as sharply after the null. Among the various single pulse phenomena exhibited by the radio pulsars, pulse nulling is the most peculiar one. Nulling pulsars are characterized by two parameters, namely, the nulling fraction (NF - which is the fraction of pulses with no detectable emission) and the nulling time scale (which is the duration for which nulling is observed). NFs are available for most of the nulling pulsars, however, the data for nulling time scale is very limited. Hence, in this work we have focused only on NFs. Nulling can also considered to be an extreme form of mode-changing phenomena [11], in which pulsar emission goes below the detection threshold.

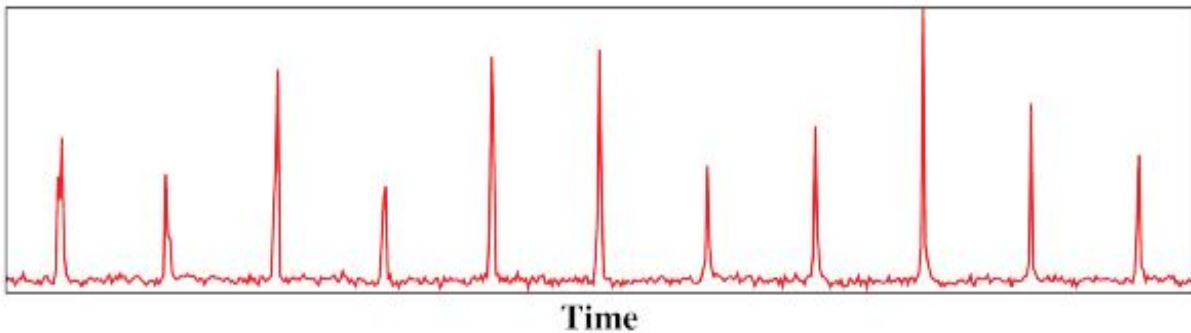


Figure 3: An observed train of pulses from PSR B0809+74 (here PSR stands for *Pulsating Source of Radio* and the number followed by it represents right-ascension and declination of the source)[4].

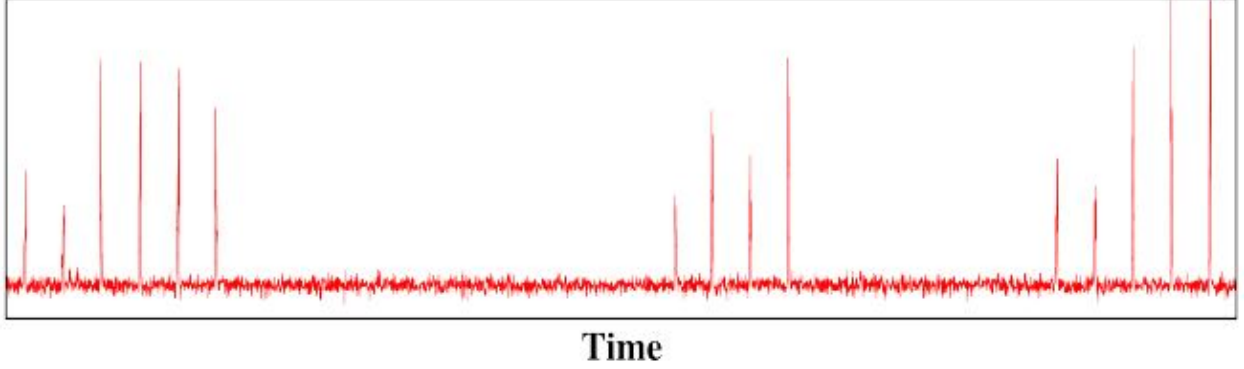


Figure 4: A sequence of pulses from PSR B2319+60 observed at 1420 MHz from the Westerbork Synthesis Radio Telescope. The nulling of pulses is clearly evident[4].

2.3 Some studies on nulling and its correlation with pulsar properties

Ritchings [12] reported correlation of NF with the pulsar period (P) and thus, concluded that the pulsars with longer period seem to possess high NFs. As the period is related to the age of the pulsar ($\propto \sqrt{P}$), he concluded that pulsars die with increasing fraction of nulling in them. Moreover, another important characteristic that he reported is that, pulsar emission is produced in burst of pulses and the length of the burst does not depend upon the age of the pulsar. However, the separations between these bursts of pulses, also known as null states, increases with the age of the pulsar. He also suggested that, if subpulse drift is due to drifting of the spark regions in the polar cap then inactive or absence of few spark regions in the observer's line-of-sight can sometime give rise to nulling in some pulsars.

A system of classification of pulsar emission phenomena has been developed in accordance with the core - cone model, which serves both as an empirical theory and as a source of insight into how individual pulsars differ from others. On the basis of their profile properties, five major categories of pulsars are delineated : the core-single (St), conal single (Sd), double (D), triple (T) and multiple/five-component (M) [13]. Rankin [14] in an attempt to correlate NF with various pulsar parameters claimed that nulling is quite common phenomena and should occur for half of the known pulsars. She also concluded that St pulsars possess small NFs compared to other classes. D and M classes possess larger number of pulsars with high NFs compared to all other classes. Studies have also been done to understand the correlation of nulling with pulsar age. It was found that in a given profile class, there is no strong correlation between the NF and the pulsar age [14]. Pulsars which possess larger NFs has no greater age than those which possess small NFs in the same class. She also pointed out that nulling is a global phenomena as it is seen in main pulse and in interpulses ². Thus, nulling represents “*global failure of the condition necessary for emission*”.

2.4 Periodic nulling

Backer [17] reported pulsar nulling in four pulsars, PSRs B0834+06, B1133+16, B1237+15 and B1929+10. He claimed the null pulses to be periodic. To find the periodicity of these null pulses, a sequence of null and burst pulses with nulled pulses as 1s and burst pulses as 0s was formed. The Fourier transform of this one-zero time series was used to look for periodicity of occurrence

²Some pulsars show two prominent components, separated by about half the period, in their integrated profiles. The more prominent component is usually called the main pulse, while the other component is called the interpulse. The main pulse and the interpulse are believed to originate from widely separated emission regions

of null pulses. Two reasons were suggested for the observed nulling: (i) the radiation region of pulsar undergoing some rapid changes. (ii) subpulse drift, as marching of subpulse with a spacing greater than the pulse profile could periodically cause null pulses. Barring the first reported study of nulling pulsars by Backer, nulling was considered a random phenomena, however, this view was challenged by later investigators. Rankin and Wright (2008) [15] concluded that an empty line-of-sight passing between the emitting sub-beams, as the cause of periodic nulling seen in PSR J1819+1305. Rankin & Wright (2007) [16] reported that nulls occur after every alternating pulses and their total observed numbers are too large for their origin from random expectation.

3 Observed properties of nulling pulsars

Since the discovery of nulling phenomena in pulsars, around 181 pulsars have been found to exhibit this behaviour. As part of our work, we did a literature survey and enlisted the nulling pulsars reported in the literature till May 2017 in Table 1 . NF values along with the basic parameters like rotation period, surface magnetic field and transverse velocities have been listed. In all of the discussion and results that follows, millisecond pulsars (MSPs -pulsars with rotation periods less than 30 milliseconds) have been excluded because of the lack of sufficient data on these MSPs.

Table 1: Table of known nulling pulsars³ with their basic parameters. The columns indicate period (P), period derivative ($\dot{P}$), surface magnetic field (B), nulling fraction (NF) and transverse velocity (V_{trans})[19]

PSRJ	P(s)	$\dot{P}(10^{15}ss^{-1})$	B($10^{12}Gauss$)	NF %	$V_{trans}(kmps)$
J0034-0721	0.942951	0.408	0.628	37.7	74.28
J0151-0635	1.464665	0.443	0.815	5	3975.42
J0152-1637	0.832742	1.3	1.05	2.5	118.54
J0229+20	0.8069	*	*	*	*
J0304+1932	1.387584	1.3	1.36	10	131.5
J0332+5434	0.71452	2.05	1.22	0.25	92.33
J0358+5413	0.156382	4.4	0.839	*	59.1
J0452-1759	0.548939	5.75	1.8	0.5	26.25
J0458-0505	1.883	0.53	1.01	63± 3	*
J0525+1115	0.354438	0.0736	0.163	0.06	264.02
J0528+2200	3.745539	40.1	12.4	25	122.56
J0529-6652	0.978	15.51	3.94	83.5± 1.5	*
J0629+2415	0.476623	2	0.987	0.02	57.64
J0630-2834	1.244419	7.12	3.01	0.3	77.29
J0659+1414	0.384891	55	4.66	12	60.3
J0726-2612	3.442	293.11	32.1	89.5	*
J0738-4042	0.37492	1.62	0.788	0.4	144.92
J0742-2822	0.166762	16.8	1.69	0.2	277.58
J0754+3231	1.442349	1.08	1.26	34	55.81
J0814+7429	1.292241	0.168	0.472	1.42	102.67
J0820-1350	1.23813	2.11	1.63	1.01	405.23
J0820-4114	0.545	0.0189	0.103	30	*
J0823+0159	0.864873	0.105	0.304	0.06	19.58
J0826+2637	0.530661	1.71	0.964	5	164.95
J0828-3417	1.848919	0.996	1.37	70	*
J0835-4510	0.089328	125	3.38	0.0008	76.97
J0837+0610	1.273768	6.8	2.98	7.1	45.98
J0837-4135	0.751624	3.54	1.65	1.2, 1.7	129.05
J0846-3533	1.116097	1.6	1.35		241.17
J0855-3331	1.268	6.321	2.86	87	*
J0922+0638	0.430627	13.6	2.45	0.05	461.12
J0934-5249	1.45	4.654	2.62	5± 3	*

³Pulsars with period greater than 30ms.

J0942-5552	0.664367	22.9	3.94	12.5	*
J0943+1631	1.087418	0.0911	0.318	8	174.47
J0944-1354	0.570264	0.0453	0.163	7	72.04
J0953+0755	0.253065	0.23	0.244	5	36.55
J1012-5830	2.134	37.65	9.07	99.6	*
J1032-5911	0.464208	3	1.19	0	*
J1049-5833	2.202325	4.41	3.15	47	*
J1055-6905	2.919	20.336	7.8	88.3	*
J1057-5226	0.197108	5.83	1.09	11	4990.7
J1059-5742	1.185	4.3	2.28	80	*
J1107-5907	0.253	0.009	0.0483	48 / 90.5	*
J1115+5030	1.65644	2.49	2.06	60	2.42E+02
J1129-53	1.063	*	*	99	*
J1133-6250	1.023	0.452	0.688	83	*
J1136+1551	1.187913	3.73	2.13	15	623.02
J1157-6224	0.4	3.9313	1.27	76	*
J1225-6035	0.626	0.288	0.43	99.4	*
J1239+2453	1.382449	0.96	1.17	6	460.32
J1243-6423	0.388481	4.5	1.34	4	*
J1255-6131	0.658	4	1.64	99.9	*
J1307-6318	4.962	21.1	10.4	90	*
J1326-5859	0.478	3.238	1.26	11.7	*
J1326-6408	0.793	3.103	1.59	87.2	*
J1326-6700	0.543009	5.31	1.72	9.1	*
J1401-6357	0.84279	16.7	3.8	1.6	*
J1406-5806	0.288	0.611	0.425	94.2	*
J1412-6145	0.315225	98.7	5.64	0	*
J1423-6953	0.333	1.4513	0.704	93.6	*
J1428-5530	0.57	2.0805	1.1	37.7	*
J1430-6623	0.785441	2.77	1.49	0.05	236.1
J1453-6413	0.18	2.7461	0.71	41	353.64
J1456-6843	0.263377	0.0983	0.163	3.3	84.34
J1457-5122	1.748	5.29	3.08	83	*
J1502-5653	0.535504	1.83	1	93	*
J1509+5531	0.74	4.998	1.95	7± 2	962.59
J1514-4834	0.455	0.925	0.656	90	*
J1525-5417	1.011694	16.2	4.09	16	*
J1527-3931	2.418	19.07	6.87	82.4	*
J1532+2745	1.124836	0.78	0.948	6	*
J1534-5334	1.368881	1.43	1.41	0.25	*
J1543-0620	0.709	0.88	0.799	4± 2	92.73
J1559-4438	0.257056	1.02	0.518	0.01	144.35
J1559-5545	0.957	20.479	4.48	96.7	*
J1607-0032	0.421816	0.306	0.364	0.1	22.8
J1614+0737	1.206801	2.36	1.71	5	*
J1624-4613	0.871	0.061	0.233	98.4	*
J1633-4453	0.436	6.2026	1.67	96.3	*
J1634-5107	0.507	1.573	0.904	86	*

J1639-4359	2.074377	2.04	2.08	0.1	*
J1644-4559	0.45506	20.1	3.06	0.4	*
J1645-0317	0.38769	1.78	0.841	0.25	189.16
J1646-6831	1.786	1.7	1.76	37.5	*
J1648-4458	0.629632	1.85	1.09	1.4	*
J1649+2533	1.015257	0.559	0.763	30	*
J1649-4349	0.872	0.044	0.198	99.4	*
J1653-3838	0.305	2.786	0.933	89.8	*
J1658-4306	1.166449	42.8	7.15	*	*
J1701-3726	2.454609	11.1	5.29	14	*
J1702-4428	2.123506	3.3	2.68	26	*
J1703-3241	1.212	0.659829	0.905	0	*
J1703-4851	1.396401	5.08	2.7	1.1	*
J1707-4729	0.266	1.5603	0.653	96.7	*
J1709-1640	0.653	6.30985	2.05	26.8	7.96
J1715-4034	0.587559	0.015	0.095	6	*
J1717-4054	0.887	3.683	1.83	>95	*
J1722-3632	0.399183	4.46	1.35	*	*
J1725-4043	1.465	2.79	2.05	97.8	*
J1727-2739	1.2931	1.1	1.21	52	*
J1731-4744	0.829829	164	11.8	0.1	2.84E+03
J1736-2457	2.642	3.43	3.05	88.2	*
J1738-2330	1.979	8.559	4.16	97.5	*
J1738-3211	0.768	0.794911	0.791	77.9	*
J1740+1311	0.80305	1.45	1.09	0.02	589.21
J1741-0840	2.043	2.27471	2.18	43.4	*
J1741-3016	1.894	8.99	4.18	97	*
J1742-4616	0.412	0.0338	0.119	99.3	*
J1744-3922	0.172444	0.00155	0.0165	75	*
J1745-3040	0.367429	10.7	2	17.5	30.82
J1749+16	2.312	*	*	*	*
J1750+07	1.909	*	*	>50	*
J1750-3157	0.91	0.19652	0.428	93.1	*
J1752+2359	0.409051	0.643	0.519	75	*
J1752-2806	0.562558	8.13	2.16	0.75	6.07
J1757-2223	0.185	0.782	0.385	98.7	*
J1758-2540	2.107	1.55	1.83	95.5	*
J1808-0813	0.876	1.2402	1.05	99.7	*
J1809-2109	0.702	3.82187	1.66	95.6	*
J1812-1718	1.205374	19.1	4.85	5.8	*
J1817-3618	0.387	2.0465	0.901	80.8	*
J1819+1305	1.060364	0.359	0.625	1	*
J1819-1458	4.263	575.171	50.1	94.7	*
J1820-0427	0.598076	6.33	1.97	0.25	20.11
J1820-0509	0.337321	0.932	0.567	67	*
J1822-2256	1.874	1.354	1.61	10± 2	*
J1823+0550	0.752907	0.227	0.418	0.4	51.06
J1823-1126	1.846	36.52	8.31	97	*

J1825-0935	0.769006	52.5	6.43	*	22.49
J1825-1446	0.279	22.67839	2.55	88.3	626.16
J1827-0750	0.27	1.5446	0.654	97.8	*
J1830-1135	6.221	47.75	17.4	79.5	*
J1831-1223	2.857941	5.47	4	4	*
J1833-1055	0.63364	0.527	0.585	7	*
J1837-0653	1.906	77.204	1.23	72.8	*
J1837-1243	1.876	0.3651	8.38	99.3	*
J1840-0840	5.309	0.237	11.3	58.4	*
J1841+0912	0.381319	1.09	0.652	5	*
J1841-0310	1.658	33.5	0.754	98.8	*
J1843-0211	2.027524	14.4	5.48	7	*
J1844+1454	0.375463	1.87	0.848	0.15	365.52
J1847-0402	0.598	51.69	5.63	3± 1	130.72
J1848-1952	4.308	23.282	10.1	27± 6	*
J1852-0635	0.524	14.64	2.8	85.9	*
J1853+0505	0.905	1.288	1.096	*	*
J1854-1557	3.453	4.52	4	*	*
J1857-1027	3.687	10.55	6.31	49.6	*
J1900-2600	0.612209	0.205	0.358	10	170.3
J1901+0413	1.978847	8.56	4.16	6	*
J1901-0906	1.782	1.638	1.73	29± 4	*
J1910+0358	2.330263	4.47	3.27	4	*
J1913-0440	0.825936	4.07	1.85	0.5	164.76
J1916+1023	1.618339	0.681	1.06	47	*
J1919+0021	1.27226	7.67	3.16	0.1	43.46
J1920+1040	2.215802	6.48	3.83	50	*
J1921+1948	0.821	0.896	0.868	8.9	*
J1921+2153	1.337302	1.35	1.36	0.25	51.54
J1926+0431	1.074078	2.46	1.64	5	*
J1926-1314	4.864	36.44	13.5	*	*
J1932+1059	0.226518	1.16	0.518	1	152.03
J1933+2421	0.81369	8.11	2.6	*	*
J1935+1616	0.358738	6	1.48	0.06	282.94
J1945-0040	1.045632	0.535	0.757	21	*
J1946+1805	0.440618	0.0241	0.104	64	12.88
J1948+3540	0.717311	7.06	2.28	0.75	457.69
J2018+2839	0.557953	0.148	0.291	0.25	31.24
J2022+2854	0.343402	1.89	0.816	3	239.01
J2022+5154	0.529197	3.06	1.29	5	107.81
J2033+0042	5.013	10.13	7.21	56± 3	*
J2037+1942	2.66308	132	18.9	26	*
J2046+1540	1.138286	0.182	0.461	0.04	211.26
J2048-1616	1.961572	11	4.69	10	510.08
J2055+3630	0.221508	0.369	0.289	0.7	63.31
J2113+4644	1.014685	0.715	0.862	12.5	186.77
J2116+1414	0.440153	0.289	0.361	1	1612.1
J2157+4017	1.525266	3.43	2.32	7.5	228.89

J2219+4754	0.538469	2.77	1.23	2	366.11
J2305+3100	1.575886	2.89	2.16	0.12	2382.31
J2317+2149	1.444653	1.05	1.24	3	*
J2321+6024	2.256488	7.04	4.03	25	2.35E+02
J2330-2005	1.643622	4.63	2.79	12	3.05E+02

4 Cause of nulling and the research problem addressed

The following figure (Fig.5) shows the distribution of non-nulling radio pulsars and nulling pulsars in the $\log P - \log \dot{P}$ plane.[21]. It is to be noted that in this plot, the pulsars that exhibit nulling are concentrated in the central region. They are not spread out like the others. It can also be found that pulsars with high NFs lie right in the middle of the normal pulsar population. Thus, it can be said that nulling does not necessarily represent dying state of a radio pulsar.

After it has been established that nulling is not due to the age of pulsar, we come across two different group of theories in literature that attempt to explain the cause of nulling. One group is more inclined towards changes in the geometry of the beam and the line-of-sight, while the second group of theories suggests intrinsic effects that can cause cessation of the full radiation beam.

4.1 Geometric Effects

It was proposed that a periodic variation in the inclination of the axis of the emission beam of the pulsar can be a likely cause for nulling. The reason for proposing this model was due to the observed periodic fluctuations seen in a few nulling pulsars. According to this model, the line-of-sight cuts between the radiating sub-beams, and thus, missing the emitting regions. It is also suggested as a possibility that only a few radiating sub-beams extinguished and when line-of-sight passes over them, nulls are observed. As the sub-beams are organized in an evenly spaced uniform pattern, empty line-of-sight cuts or extinguished individual sub-beams are more likely to repeat after certain pulsar periods. This was attributed as the periodicity observed in nulling. [4]

There are other models of geometric origin too, such as the one in which it was reported that the precession of the rotation axis is the cause of nulling. However, the precession in pulsars is rare and there are not many tentative evidence. Others have reported deformation of the emission beam due to the misalignment between the rotation and the neutron star symmetry axis. [4]

4.2 Intrinsic effects

We know that the coherent radiation is a highly tuned process. Emission may stop if there is any change in the necessary conditions of coherence. Cheng [22] suggested fluctuations in the temperature to invoke changes in the coherence condition as a possible cause of nulling. In another model suggested by Filippenko and Radhakrishnan [23], they proposed continuous flow of charged particles as the cause of nulling. According to Kazbegi et al.[24], the drifting of sub-pulses are due to rotation of plasma wave around the magnetosphere and not due to the shifting spark regions. The chemical composition and structure of the neutron star also plays an important role in the characterization of its emission behavior. There are suggestions that it is possible for the neutron star surface to have irregularities of the order of around $10cm$. These tiny hills on the polar cap region can heavily affect the outflow of particles. This model can be extrapolated to suggest that it can also cause the radio emission to go below the detection threshold of the telescope to produce nulls.[4]

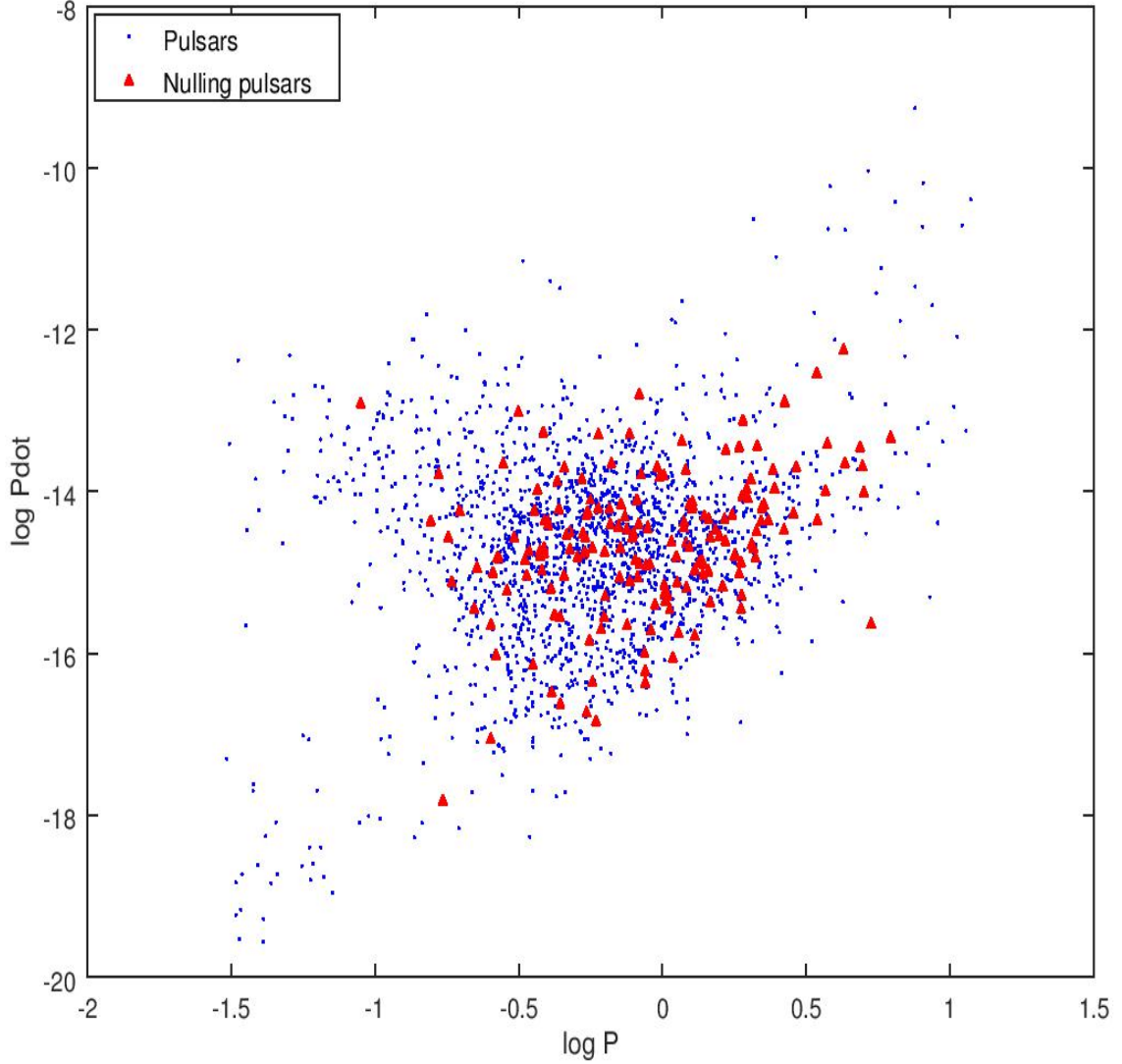


Figure 5: $\log P$ vs $\log \dot{P}$ graph for normal radio pulsars and nulling pulsars.

4.3 External effects

Apart from the two effects mentioned earlier, some models also propose theories which are related to external effects - such as magnetospheric changes - causing nulling. It was suggested by Michel [25] that nulling might be due to accretion of interstellar medium to pulsar magnetosphere. In another study by Cordes & Shannon [26], it was proposed that influence of asteroid belt around the pulsars caused various temporal changes. This cirumpulsar debris entering the light cylinder can change the conditions for pair productions and coherence, which can cause cessation of observed emission.

4.4 Motivation for this study and the problem addressed

Pulsar nulling remains an unexplained phenomena till date. The fundamental question still remains - *Why does some pulsars null and others do not?*. Models proposed in this regard aim to tackle the problem for individual cases. A lack of common agreement between different studies regarding the correlations of NFs with different pulsar parameters might have prevented a satisfactory explanation on the broader picture. In this work we have tried to address the aspect of nulling in a broad overall picture by comparing the properties of nulling pulsars to that of the normal radio pulsars and checking for any observable and significant trend. Specifically, we try to search for correlation between nulling in pulsars and the observed transverse velocities (V_{trans}).

5 Proper motions and pulsar kicks

5.1 Proper motion

Proper motion is defined as the observed changes in the apparent location of astronomical objects (in this case, pulsars) with respect to the distant background. The closest stars have large enough proper motions to be expressed in arcseconds per year, although milliarcseconds per year is more common. It requires more than two epochs to be able to separate out the proper motion of a star from its parallax, as both cause the stars to move against the fixed background of the night sky. However, it may take several epochs to separate the two terms due to experimental uncertainties. The proper motion (μ) has two components - along the right ascension (μ_{RA}) and along the declination (μ_{Dec}).

$$\mu^2 = \mu_{RA}^2 + \mu_{Dec}^2$$

The transverse velocity (V_{trans}) is the velocity perpendicular to our line of sight. It is related to the proper motion as:

$$V_{trans} = \mu \times D$$

where D is the distance to the star⁴. Combining this with the radial velocity gives the 3D space velocity of the star.

The proper motions of a number of pulsars are large enough to be measured by observing pulse arrival times over a period of several years. Manchester et al. [27] used timing observations over a four-year period for PSR B1133+16 and determined the proper motion for the first time. However, this method of measuring proper motion has proved to be difficult and unreliable because of the random timing noise observed in most pulsars. Most of the known values of proper motion have been found through direct interferometric measurements.

⁴Check Appendix A for the ways to measure distances of pulsars

5.2 Graphical analysis of the available data on transverse velocities for nulling and non-nulling pulsars

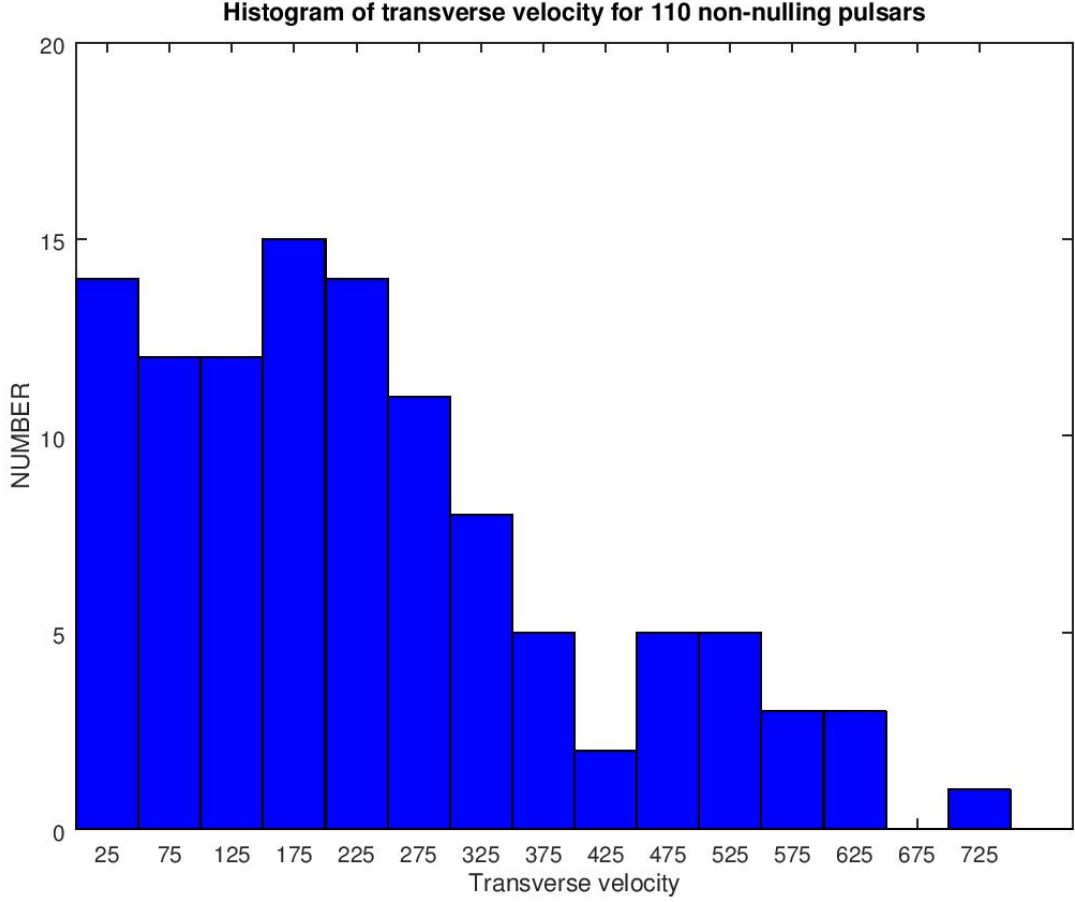


Figure 6: Histogram of V_{trans} (in km/s) for 110 non-nulling pulsars⁵ with bin size 50 km/s

Figs. 6 and 7 clearly indicates that the average speed for nulling pulsars is less than that of the non-nulling ones. Pulsars with higher transverse velocities are found at a greater distance away from the Galactic plane. So, this could mean that nulling pulsars are seen frequently near the Galactic plane. This could be explained as follows: near the galactic plane, the density of ionized gas clouds are larger, and hence nulling could be caused by a randomly moving ionized gas cloud eclipsing the pulsar for some time.

⁵Out of a total of 2084 non-nulling normal radio pulsars, V_{trans} values are available for 110 pulsars.[19]

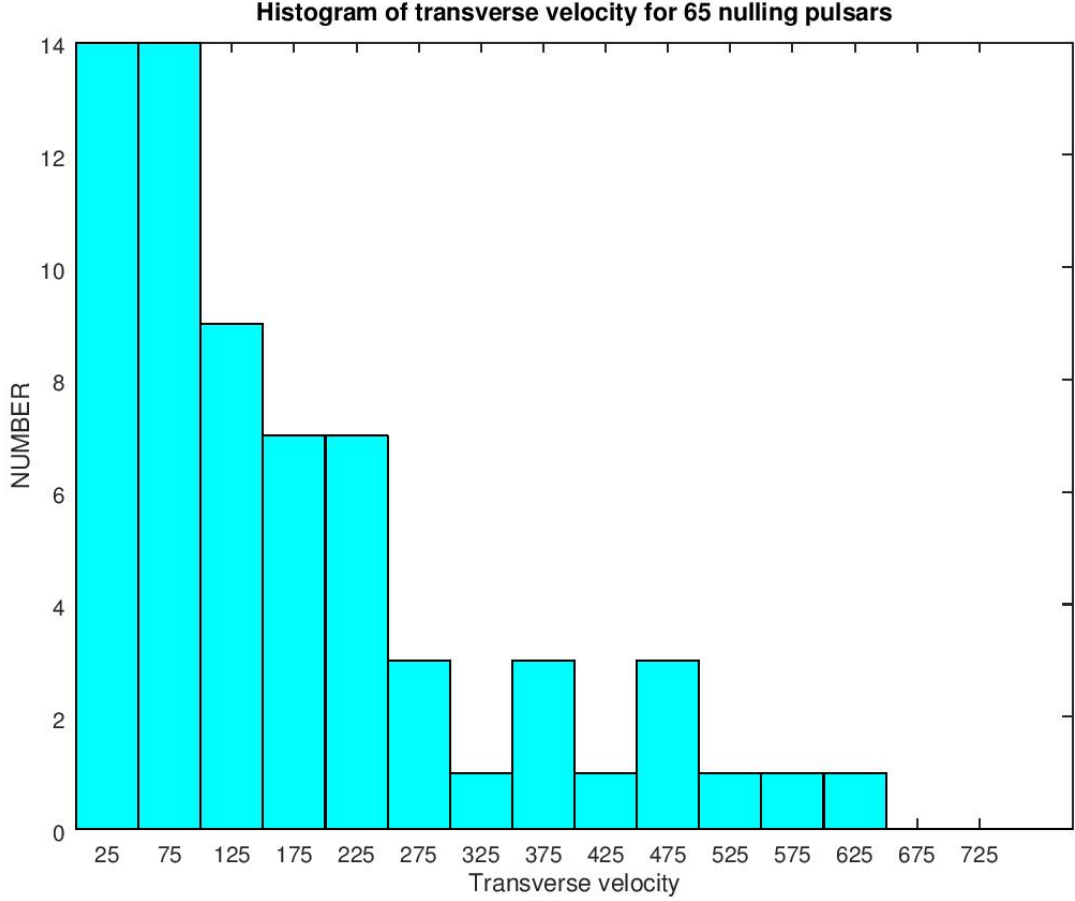


Figure 7: Histogram of V_{trans} (in km/s) for 65 nulling pulsars⁶ with bin size 50 km/s

5.3 Pulsar birth kicks

It was long understood that neutron stars have space velocities much greater than those of their progenitors. Studies of pulsar proper motion give 200–500 km/s as the mean 3D velocity of neutron stars at birth and a significant population having velocities greater than 1000 km/s [29]. Such high velocities can be explained by considering that the supernova explosions are asymmetric and provide kicks to nascent neutron stars. Neutron star kicks are supported by observations of orbital plane precession in PSR J0045-7319 and its fast orbital decay and the high eccentricities of X-ray binaries. Study of evolution of neutron star binary population also imply the existence of pulsar kicks. Asymmetry in supernova explosions is also directly observed by studying nearby supernovae and supernova remnants.

Theories propose that such natal kicks can be generated by an asymmetric explosion due to global hydrodynamic perturbations in the supernova core or due to asymmetric neutrino emission in the presence of super-strong magnetic fields ($B > 10^{15}G$) in the proto-neutron star. An alternative theory on post-natal kick mechanism relies on asymmetric electromagnetic radiation from an off-centered dipole in a rapidly rotating pulsar. Since there is a lack of simple relationship between

⁶Out of a total of 181 nulling normal radio pulsars, V_{trans} values are available for 65 pulsars.[19]

velocity and other properties of neutron stars, hence it is difficult to determine the exact mechanism at play.

5.4 Neutrino emission driven kicks

We now make a simple analysis to check if the asymmetric emission of neutrinos can have a significant contribution on the collapsing core of the progenitor.

We treat neutrinos as massless. So, momentum,

$$p = E_\nu/c$$

where E_ν = total energy of the particle, c = speed of light.

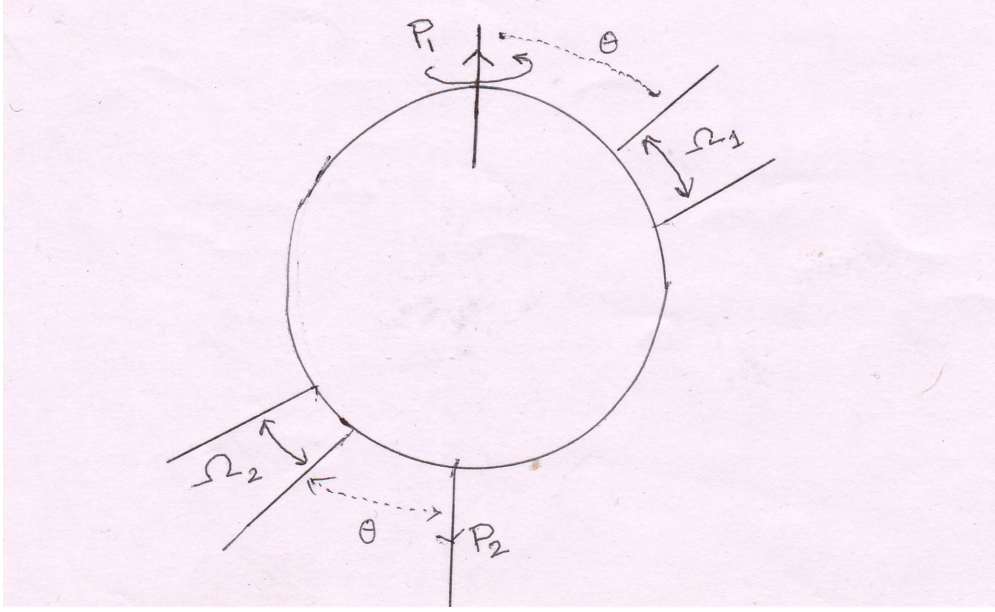


Figure 8: Neutrino emission from pulsars

Consider asymmetric neutrino emission from pulsars at an angle θ . Let P_1 and P_2 be the momentum components in the upward and downward components respectively. Let Ω_1 and Ω_2 be the beam angles as shown in fig. 8. We assume that the energy emitted is uniform.

$$P_1 = \frac{E_\nu}{4\pi c} \Omega_1 \cos \theta$$

$$P_2 = \frac{E_\nu}{4\pi c} \Omega_2 \cos \theta$$

$\therefore$ Net momentum carried away by the neutrinos in the upward direction

$$= \frac{E_\nu \cos \theta}{4\pi c} (\Omega_1 - \Omega_2)$$

Hence, recoil momentum of the neutron star is given by,

$$M_{NS} \cdot v_k \approx \frac{E_\nu \cos \theta}{4\pi c} (\Omega_2 - \Omega_1)$$

where, M_{NS} is the mass of the neutron star and v_k is the kick velocity gained by the neutron star

$$\therefore v_k \approx \frac{E_\nu \cos \theta}{4\pi c \cdot M_{NS}} (\Omega_2 - \Omega_1)$$

For a back of the envelope calculation, we plug in some typical numbers:

$$E_\nu \sim 10^{53} \text{ ergs}$$

$$M_{NS} \sim 1.4 \times 2 \times 10^{33} \text{ g}$$

$$c = 3 \times 10^{10} \text{ cm/s}$$

which gives,

$$\begin{aligned} v_k &\approx \frac{\cos \theta (\Omega_2 - \Omega_1) \cdot 10^{53}}{4\pi \times 3 \times 10^{10} \times 1.4 \times 2 \times 10^{33}} \\ &\approx 10^4 \cdot \cos \theta \frac{\Omega_2 - \Omega_1}{4\pi} \text{ km/s} \end{aligned}$$

Taking $\cos \theta = 1/2$ and $(\frac{\Omega_2 - \Omega_1}{4\pi}) = 0.01$, we get

$$v_k = 50 \text{ km/s}$$

Thus, this tells us that indeed asymmetric emission of neutrinos **can** give large kick velocities.

6 Study of correlation between V_{trans} and NF for nulling pulsars

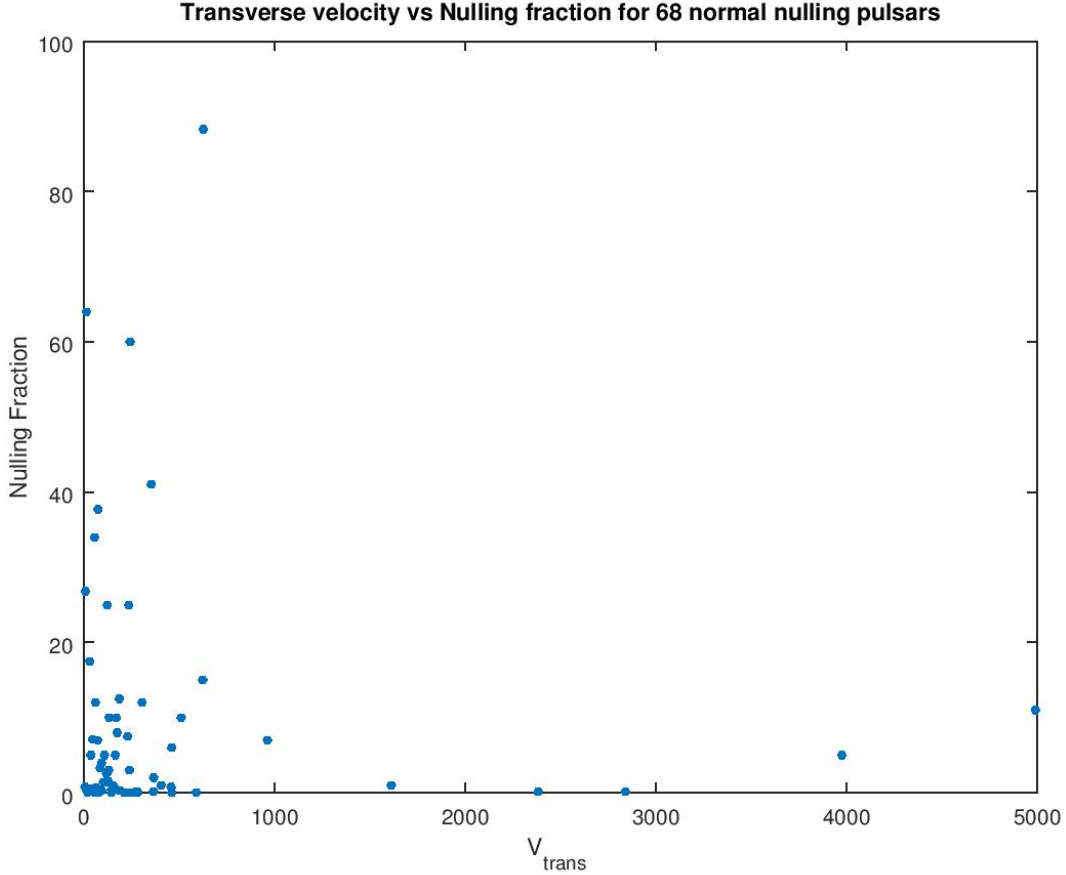


Figure 9: Distribution of V_{trans} vs NF for 68 nulling pulsars

Correlation refers to a broad class of statistical relationships involving dependence, though in common usage it most often refers to how close two variables are to having a linear relationship with each other. Formally, if random variables do not satisfy the mathematical property of *probabilistic independence*, they are termed dependent. Statistically, correlation establishes a relationship between mean values.

Out of the several correlation coefficients, the most commonly used one to measure the strength of a linear association between two variables is known as the Pearson correlation coefficient. The value of this coefficient lies between +1 and -1, where the value $r = 1$ means a perfect positive correlation and the value $r = -1$ means a perfect negative correlation. When the correlation coefficient is close to zero there exists no evidence of any relationship. However, it is to be noted that “*Correlation does not imply causation*”.

To determine the Pearson's correlation coefficient, the data sets must satisfy the following requirements[31]

- The distribution of variables must be approximately normal distribution.
- The scale of measurement should be interval or ratio.

- Both data sets should be linearly associated.
- Data should not have any outliers.

If x_i and y_i represent the i^{th} values in the two data sets respectively, and if $\bar{x}$ and $\bar{y}$ represent the mean values of data in the two sets, then the Pearson's correlation coefficient, r , can be determined from the following equation:

$$r = \frac{\sum_i (x_i - \bar{x})(y_i - \bar{y})}{\sqrt{\sum_i (x_i - \bar{x})^2} \sqrt{\sum_i (y_i - \bar{y})^2}}$$

It is not just by the correlation coefficient that confidence in a relationship is formally determined, but also by the number of pairs in the data. If there are very few pairs then the coefficient will be deemed ‘statistically significant’ when it is very close to +1 or -1, but if there are many pairs then a coefficient closer to 0 can still be considered ‘highly significant’. The ‘significance’ of this empirical analyses is measured by the p-value. To calculate the p-value, we start with the *null hypothesis*, which in this case is the statement that “ V_{trans} and NF of nulling radio pulsars are unrelated”. P-value lies between 0 and 1 indicating the probability that this data would have arisen if the null hypothesis were true. A low p-value tells us that the null hypothesis can be rejected.

The Pearson's correlation coefficient r in our study has been calculated using software[21] and turns out to be:

$$r = -0.0306$$

The p-value has been calculated online[32],

$$p - value = 0.80185$$

7 Summary

In this work we did a literature survey of the nulling pulsars detected till May 2017 and included properties like transverse velocity, V_{trans} , and Nulling Fraction, NF, for our analysis (The transverse velocity values were obtained with respect to the galactic coordinate system). By analyzing the data, we can find that average V_{trans} of the nulling pulsars is less than that of the non-nulling pulsars, which made us propose that nulling pulsars could be found close to the galactic plane. We also propose that nulling could be due to ionized gas clouds. For a quantitative study of the same, we did a correlation check between NF and V_{trans} and found the Pearson's correlation coefficient for these two data sets to be -0.0306 with a p-value of 0.80185.

Previous works on the pulsar nulling phenomena have been studied along with the proposed theories. Pre-natal kick to explain the high velocity of pulsars have been the major motivation for our work. Neutrino emission proposed as a driving mechanism for this kick has been a part of our concentration in this work.

Future studies concerned with our work can incorporate data for the Millisecond Pulsars (MSPs), since data for MSPs are insufficient as of now.

In order to develop a theory of the mechanisms that the pulsars exhibit, with nulling in particular, we are dependent on the available data. More the data, better results can be expected out of the theoretical model. Studies like ours can be pivotal in understanding the formation and evolution mechanisms of pulsars.

Appendices

A Distances of the pulsars

The distances of nearer stars can be determined from parallax measurements, which is the apparent annual cyclic movement of position due to the Earth's orbital motion around the Sun. For distances greater than about *1kiloparsec*, apparent brightness can be used to infer the stellar distances, since the absolute brightness of a star is generally available from its spectral type. For pulsars, the intrinsic brightness is a variable quantity, in contrast to that of a visible star of known type. The intrinsic brightness varies from time to time and from pulsar to pulsar. So an observed intensity gives little indication of distance.

In pulsar astronomy, dispersion measure (DM) manifests itself observationally as a broadening of an otherwise sharp pulse when a pulsar is observed over a finite bandwidth. DM serves as a remarkable direct measurement of distance in the frequency dispersion due to the passage of the radio pulses through the ionized interstellar medium. Technically, DM is the *integrated column density of electrons along the line of sight*. In simple terms, DM represents the number of free electrons between us and the pulsar per unit area. Hence, the interpretation of measurements of DM of pulsars as distance requires a model of the electron distribution in interstellar space; this is itself largely constructed from the measured values of DMs of pulsars whose distances are known by parallax measurements, or by absorption in neutral hydrogen, or by association with optically identified objects[20].

In the presence of charged particles, the electrostatic interaction between light and charged particles causes a delay in the propagation of light. This delay is a function of the frequency of the radio wave and the masses of the charged particles. Since radio photons have low energy so they are more significantly delayed. Similarly, electrons, due to their much lower masses in comparison to the protons, react to the passing light more and cause greater delays in the light propagation time. The net result is a delay in the arrival of pulses and the amount of delay depends upon the radio frequency and the DM.

The amount of DM smearing τ_{DM} across a finite bandwidth $BMHz$ at a central frequency of νGHz is proportional to the DM and given by[28]:

$$\tau_{DM} = 8.3 \times B \times DM \times \nu^{-3} \mu s$$

The time delay $t_2 - t_1$ between two observing frequencies ν_1 (in GHz) and ν_2 (in GHz) is:

$$t_2 - t_1 = 4.15 \times DM \times [\nu_1^{-2} - \nu_2^{-2}] ms$$

From this we can compute the DM by measuring the time delay between two frequency bands.

A model of the galaxy's free electron density (n_e) along with the DM can serve as a distance(D) indicator:

$$D = \frac{DM}{n_e} \text{ parsec}$$

where DM is in parsec cm^{-3} and n_e is in cm^{-3} .

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